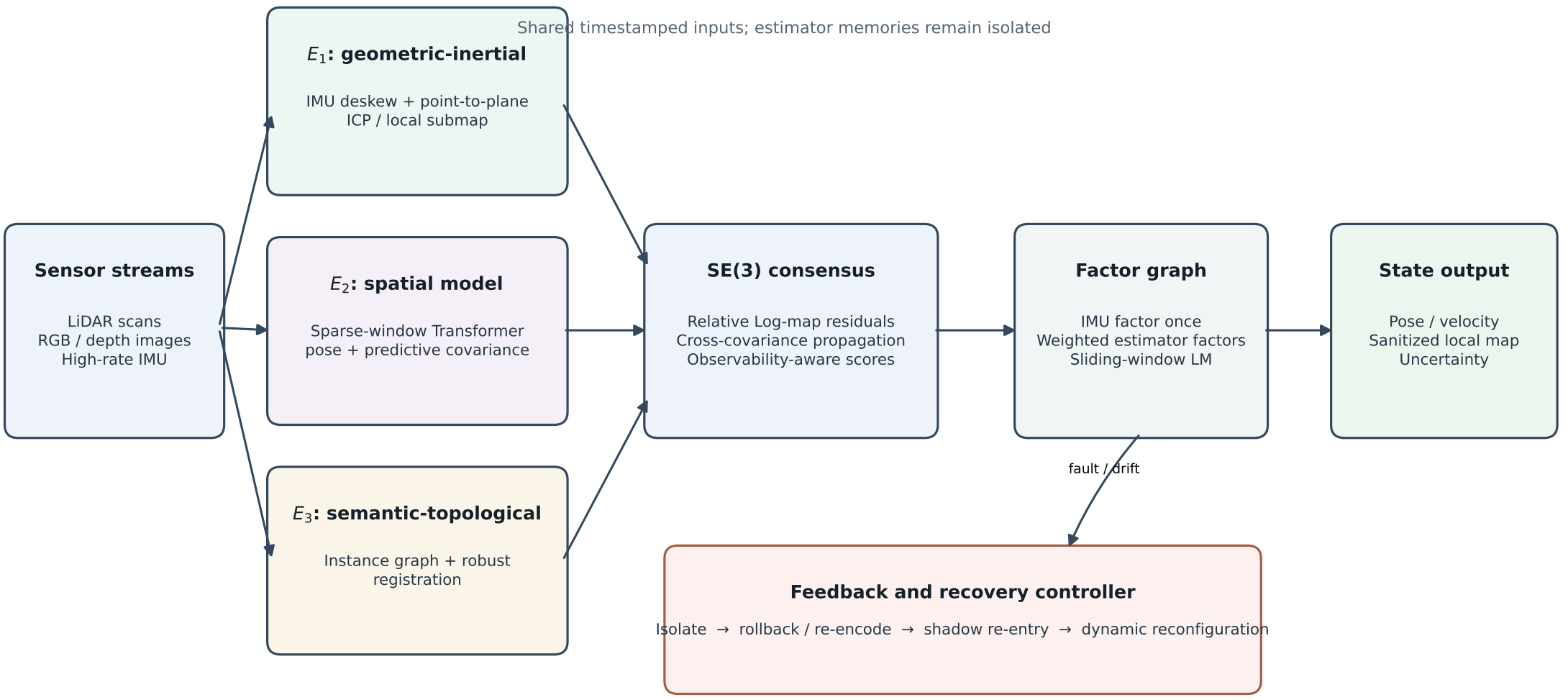
EndoSpatial-SLAM: heterogeneous estimation and feedback recovery
Shared timestamped inputs; estimator memories remain isolated
Sensor streams
LiDAR scans
RGB / depth images
High-rate IMU
$E_1$: geometric-inertial
IMU deskew + point-to-plane
ICP / local submap
$E_2$: spatial model
Sparse-window Transformer
pose + predictive covariance
$E_3$: semantic-topological
Instance graph + robust
registration
SE(3) consensus
Relative Log-map residuals
Cross-covariance propagation
Observability-aware scores
Factor graph
IMU factor once
Weighted estimator factors
Sliding-window LM
State output
Pose / velocity
Sanitized local map
Uncertainty
fault / drift
Feedback and recovery controller
Isolate → rollback / re-encode → shadow re-entry → dynamic reconfiguration